Operations Management

Session 3: Service Oriented Operations

Outline:

- Service Processes
  - The Law of Service & the Gap Model
  - Managing Intangibility, Perishability, Heterogeneity
- Shouldice
  - Focus Oriented Operations
  - Monetizing the Ops Quadrangle
  - Customer-Operator opportunities
- Perishable Asset Revenue Management (PARM)
  - Synchronizing Supply with Demand in Real Time
  - Managing Revenue per Available Resource (RevPAR)
  - Bid Price Controls

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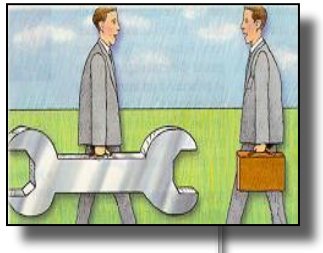
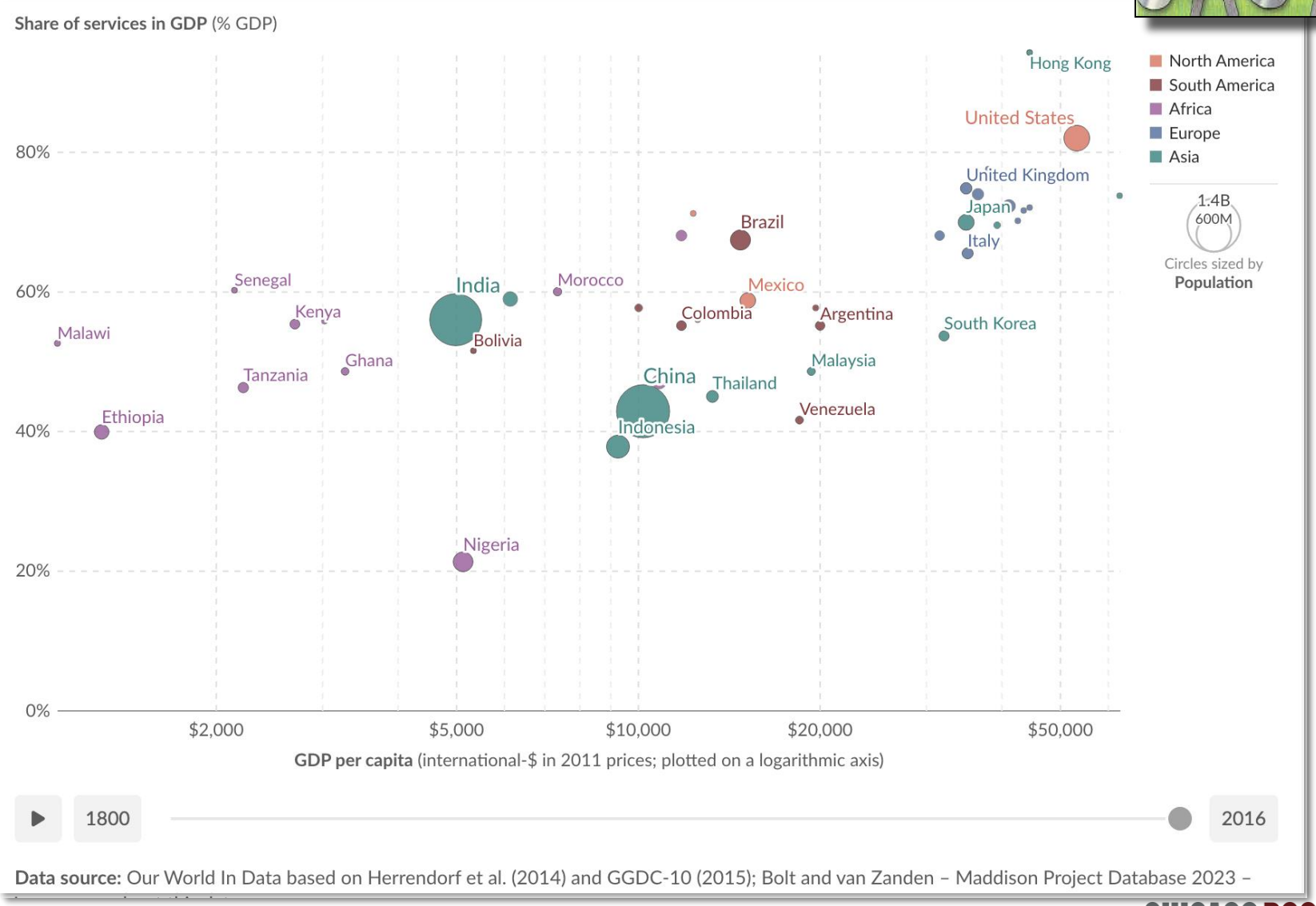
The Economy of Services





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How Have We Done?

Services: With a Smile ☺

Percentage of consumers who rated service either excellent or very good for the following businesses in the USA

Business	Percentage
Overnight Mail	43
Long-distance Phone	38
Local Telephone	37
U.S. Postal Service	37
Doctors/Health Care	36
Supermarkets	36
Domestic Air Travel	34
International Air Travel	29
Banks	29
Full-Service Restaurants	29

Business	Percentage
Health Insurance Firms	25
Hotels/Motels	25
On-line Services	25
Department Stores	25
Discount Stores	25
Catalogue/Mail Order	20
Public Transportation	19
Investment Brokers	17
Fast-Food Restaurants	17
Automobile Dealers	14

Source: New York Times.

The Economy of Services

Labor Intensity of Service Groups

Industry	Labor-Capital Ratio
Electric Utilities	0.07
Communications	0.19
Hospitals	0.61
Auto & Other Repair	0.63
Transportation	0.79
Banking	0.83
Hotels	0.99
Retail Trade	1.61
Advertising	2.38
Insurance	5.56
Brokerage	6.67

Increasing Labor Intensity

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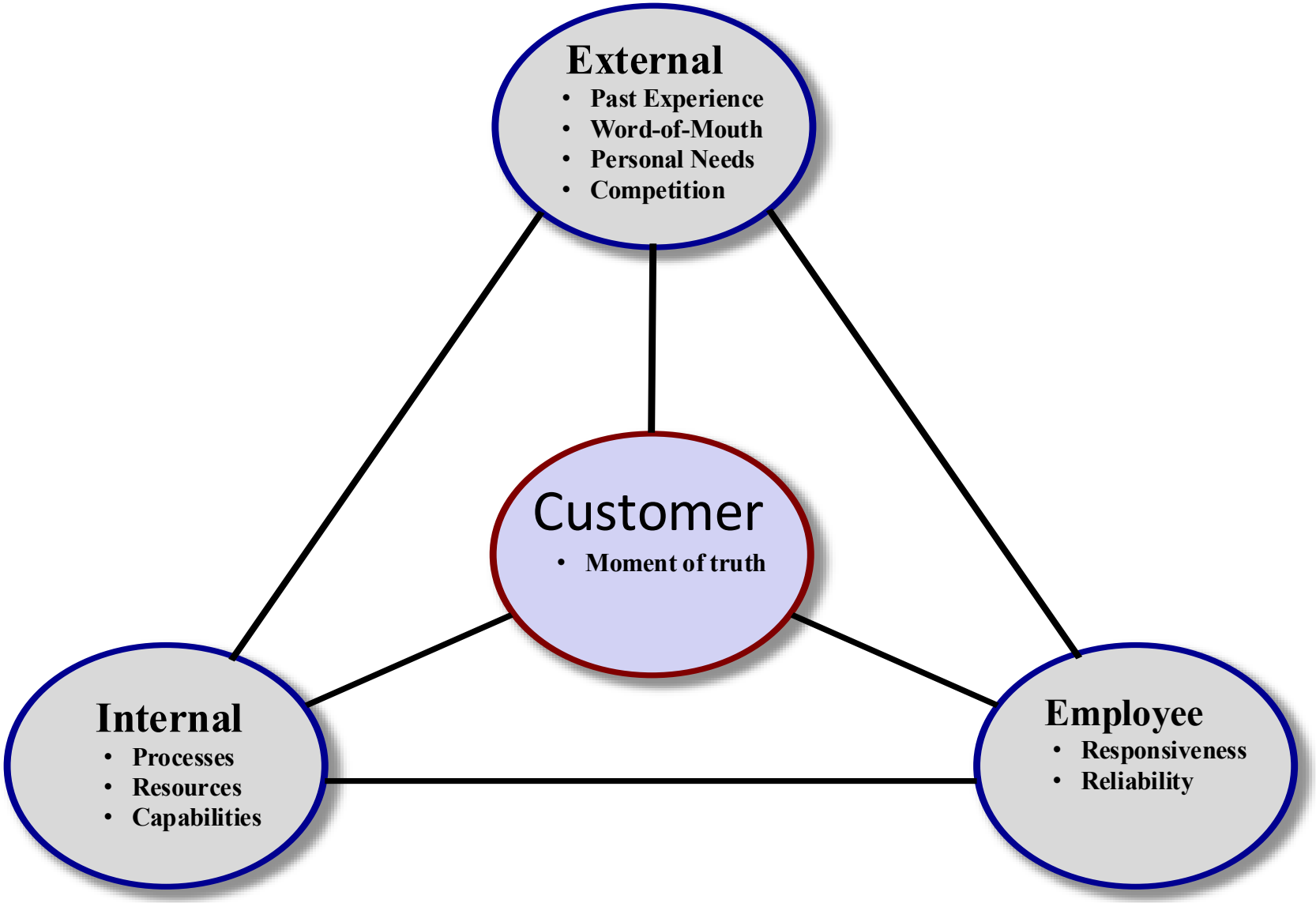
A Framework for Services



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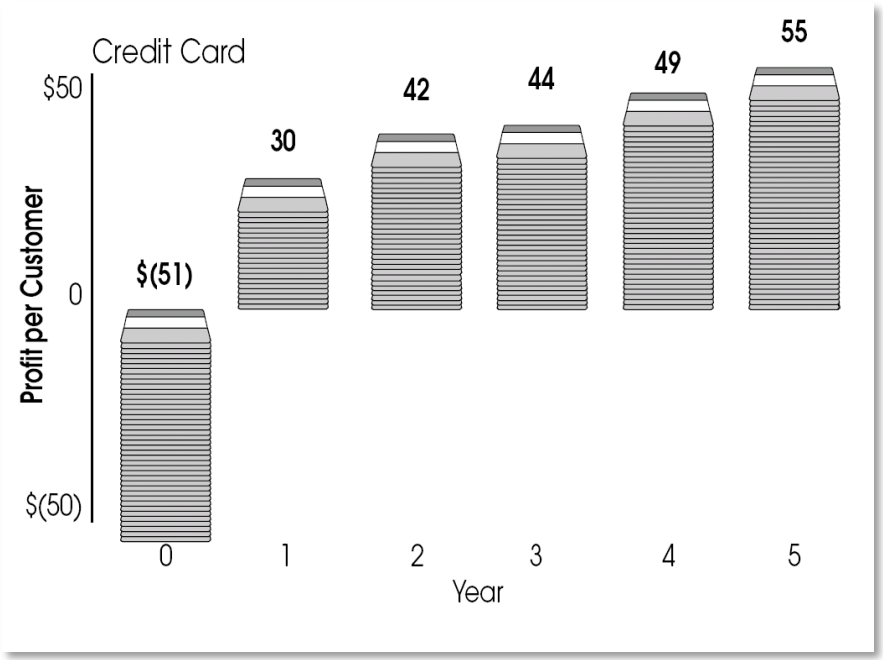
The Fundamental Law of Service Management

Customer Satisfaction = Perception - Expectation



How Much Profit Satisfied Customers Generate over Time

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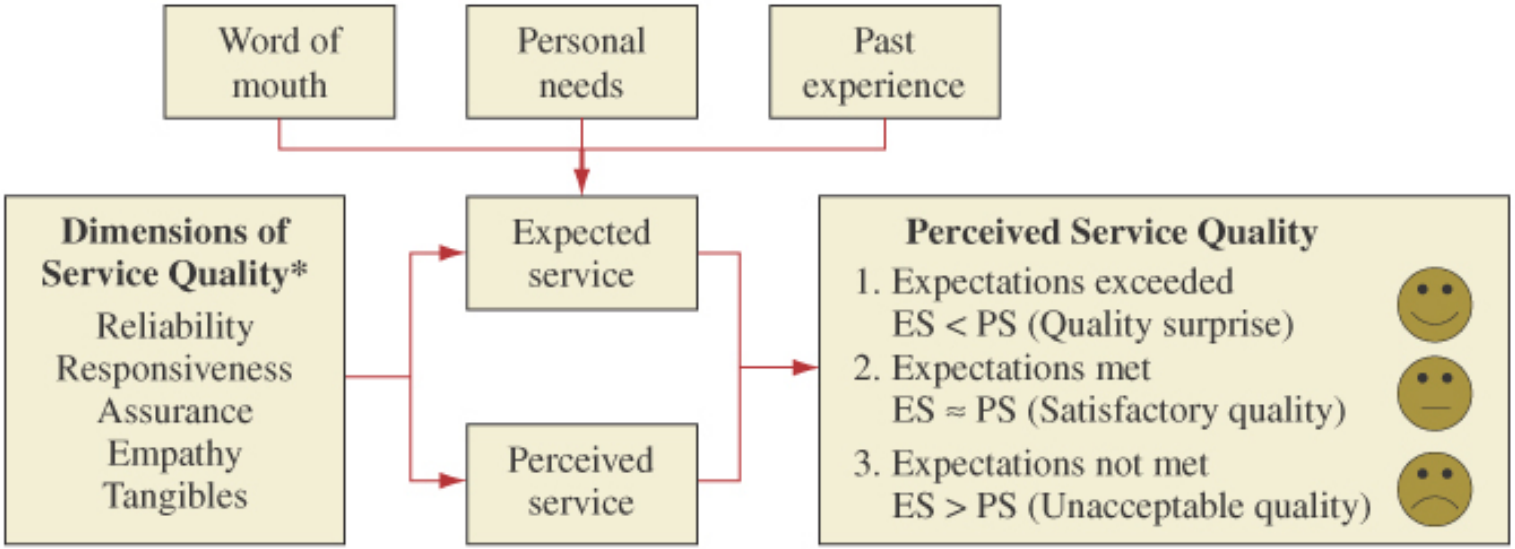
Measuring Satisfaction: The Gaps' Model

Perceived Service Quality



**Reliability:* the ability to perform service as promised, both dependably and accurately.

Responsiveness: willingness to help customers promptly.

Assurance: knowledge and courtesy of employees, as well as their ability to convey trust.

Empathy: caring and individualized attention.

Tangibles: the appearance of physical facilities, equipment, and personnel, as well as other factors affecting the senses such as noise and temperature.

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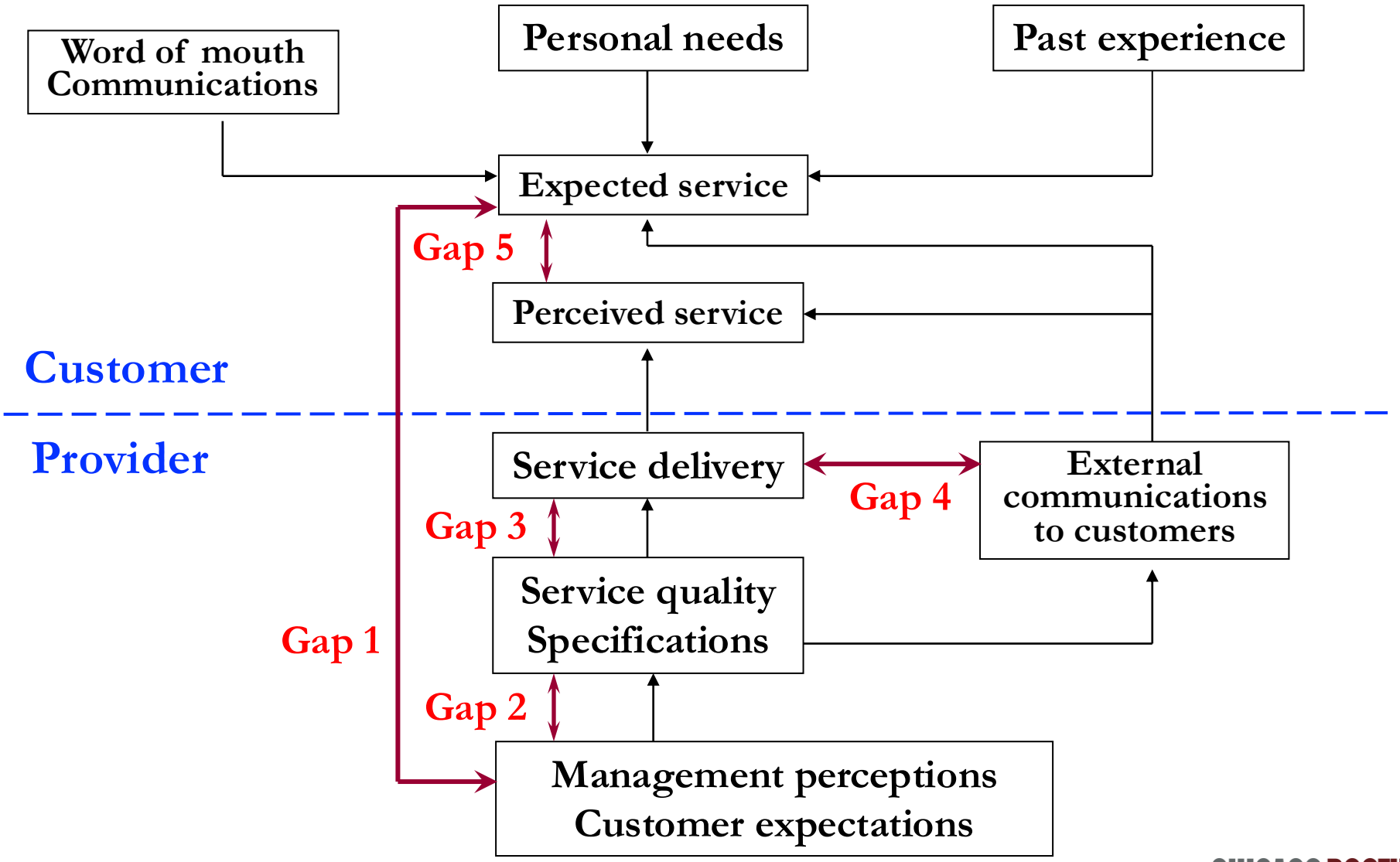
Measuring Satisfaction: The Gaps' Model



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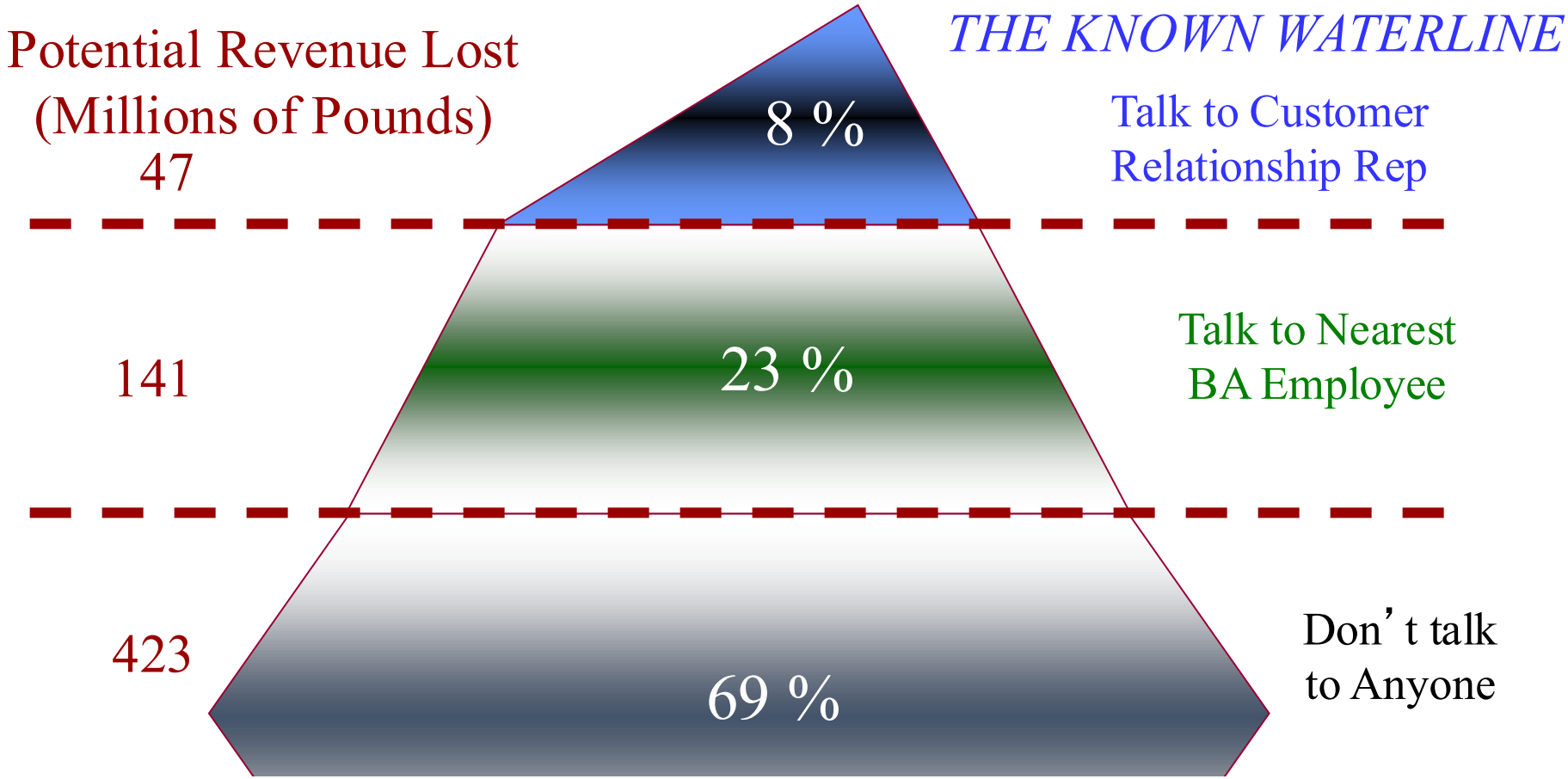
Measuring Gaps

THE "COMPLAINANT ICEBERG" AT BRITISH AIRWAYS



An opportunity occurs somewhere on the network every 15 seconds (1/3 of total customers)

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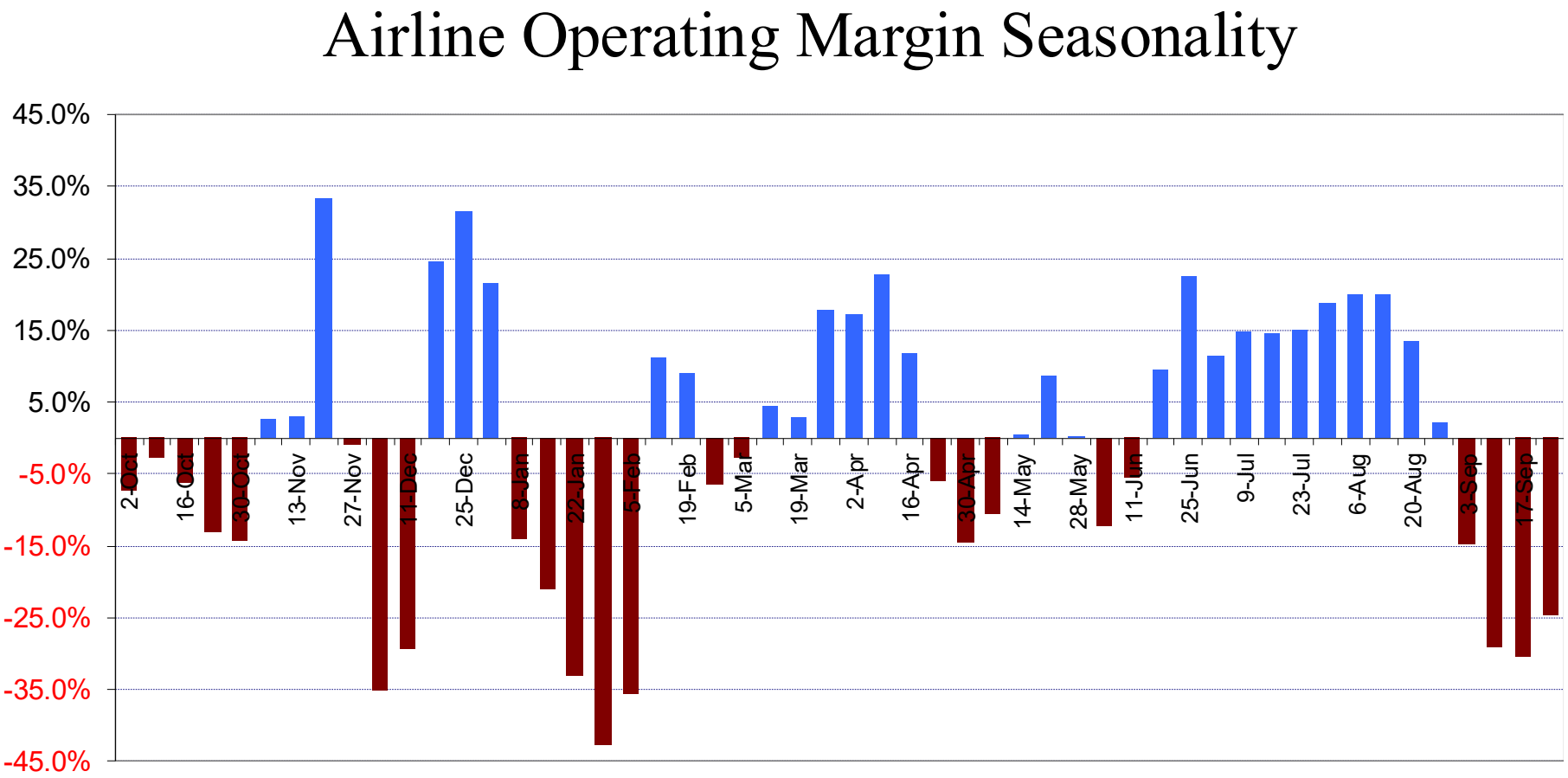
Why Services are Hard to Manage?

- Intangibility
  - Appropriability
  - Service Materialization
- Perishability
  - Matching Demand and Supply & Variability
- Heterogeneity
  - Standardization, Training, Segmentation
- Simultaneity & Transferability

Perishability



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Heterogeneity

The Production Line Approach

The Self-Service Approach

The Personal Attention Approach

Operational Strategies

Matching supply and demand is a critical step in the service industry where traditional inventory techniques are not feasible.

	Capacity Management	Demand Management
	Location	Pre-selling
	Modular facility design	Direct marketing
	Sharing capacity	Price differentiation
	Technology	Information to customers
	Information systems	Promotion and sales
Tactical	Standardization	Complementary services
Level	Floating staff and part-time employees	Preventive maintenance
	Cross-training	of user's equipment
	Extending business hours	
	Preprocessing	
	Preventive maintenance	
	Cooperative with competitors	
	Complementary Services	
	Downgrading of products	Pricing
	Overbooking	Loss leaders
Operational	Service length	
Level	Scheduling	
	Engaging customers	
	Batching the service delivery	

(Extracted from Bitran G. and Mondschein S. (1997))

Shouldice

SHOULDICE SURGERY
ESTABLISHED 1945
Shouldice
HOSPITAL
Shouldice at a Glance
About Us
The Centre of Excellence for Hernia Repair
The global leader in external abdominal wall hernia surgery for over 65 years
Arjuna Rousseau, Patient, Costa Rica
Click here to arrange an exam

Shouldice: Operating System

1. How successful is Shouldice? How do you know?
2. How is quality controlled?
3. Given that there is almost zero "repurchased", what incentives does Shouldice have to satisfy patients?
4. How does Shouldice lower its operating cost?
5. How does Shouldice maximize perceived customer value?
6. How does Shouldice manage Perishability and Heterogeneity?

Shouldice: Operating System

CAPABILITIES = F(RESOURCES, PROCESSES, OPERATIONS CULTURE)

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Shouldice: Value Creation

Focused Process (Factory)!

	SHOULDICE	REGULAR HOSPITAL
PRICE		$5,240
QUALITY		10% recur.
TIME		5-8 day stay 2-8 week recovery
VARIETY	Hernias	

ARAVIND EYE CARE SYSTEM

YouTube [Link]

Shouldice Capacity Analysis

Options for expansion:

- New 45 bed wing
- Operate on Saturday and/or Sunday

Analysis of expansion:

- Like to build a tool, or model, to understand impact of various options
- First, how does Shouldice generate revenue?
- Second, how do various options affect revenue?

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Shouldice Revenues

HOSPITAL:

CLINIC:

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How does Shouldice make money?

Shouldice Capacity Analysis

BEDS:

OPERATING ROOMS:

SURGERY TEAMS:

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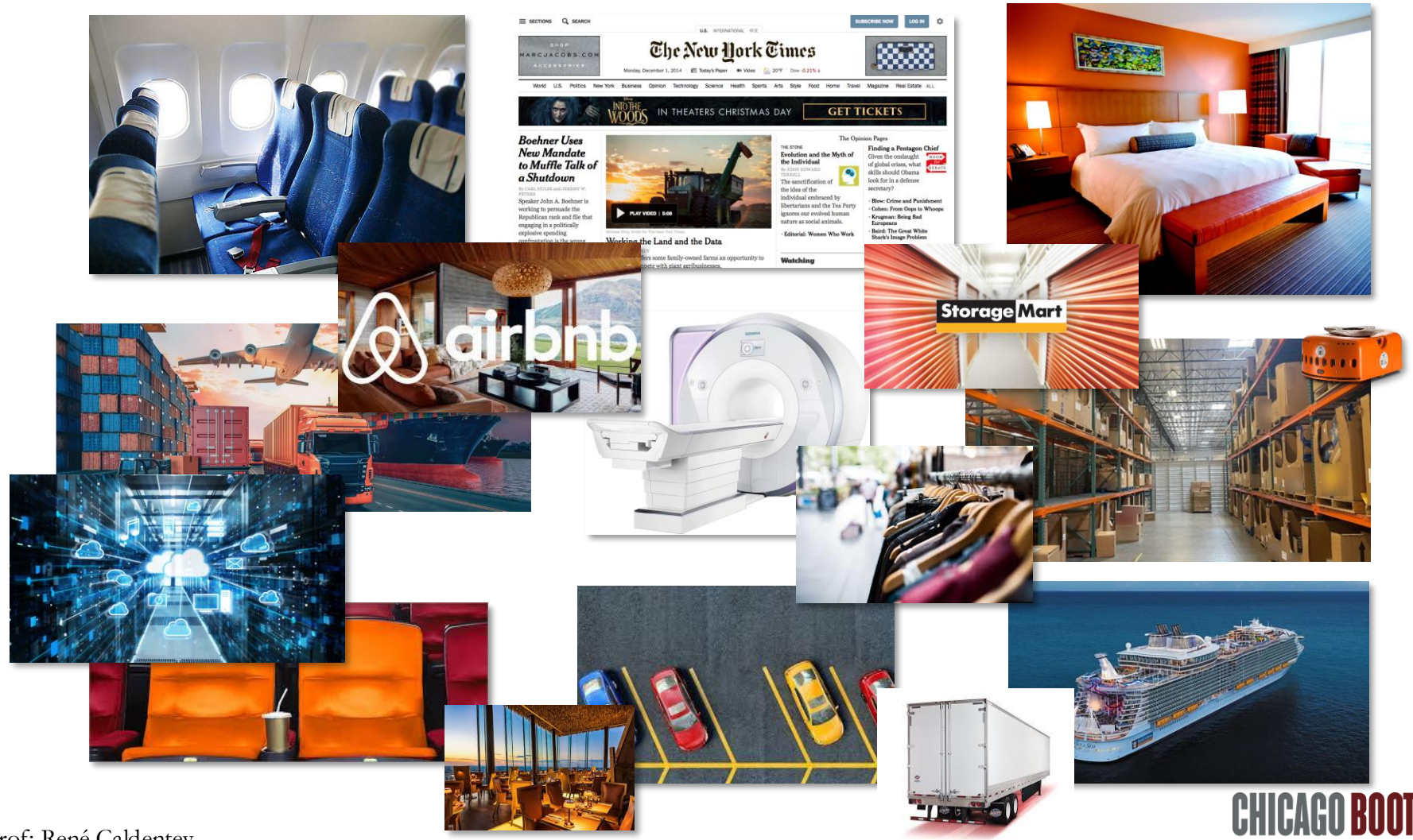
Perishable Asset Revenue Management (PARM)



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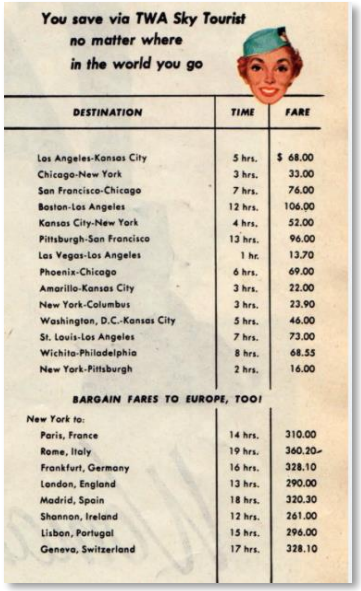
Airline Industry Origins



You save via TWA Sky Tourist no matter where in the world you go

DESTINATION	TIME	FARE
Los Angeles-Kansas City	5 hrs.	$ 68.00
Chicago-New York	3 hrs.	33.00
San Francisco-Chicago	7 hrs.	76.00
Boston-Los Angeles	12 hrs.	106.00
Kansas City-New York	4 hrs.	52.00
Pittsburgh-San Francisco	13 hrs.	96.00
Las Vegas-Los Angeles	1 hr.	13.70
Phoenix-Chicago	6 hrs.	69.00
Amarillo-Kansas City	3 hrs.	22.00
New York-Columbus	3 hrs.	23.90
Washington, D.C.-Kansas City	5 hrs.	46.00
St. Louis-Los Angeles	7 hrs.	73.00
Wichita-Philadelphia	8 hrs.	68.55
New York-Pittsburgh	2 hrs.	16.00
BARGAIN FARES TO EUROPE, TOO!		
New York to:		
Paris, France	14 hrs.	310.00
Rome, Italy	19 hrs.	360.20
Frankfurt, Germany	16 hrs.	328.10
London, England	13 hrs.	290.00
Madrid, Spain	18 hrs.	320.30
Shannon, Ireland	12 hrs.	261.00
Lisbon, Portugal	15 hrs.	296.00
Geneva, Switzerland	17 hrs.	328.10

Standardized Prices and Profitability Targets

Low-Cost Airlines (PeopleExpress)

Computerized Reservation Systems (CRS)

Global Distribution Systems (GDS)

1986 PeopleExpress went bankrupt

Adoption of RM by Major Airlines (Continental, Delta, United)

Expansion
- Hotels
- Pass. Railroads
- Car Rental
- Cruise Lines
- Shipping
- Trucking
- Broadcasting
- Retail

1970

1978
Airline Deregulation Act

1985
American Airlines' "Ultimate Super Saver Fares" & DINAMO

1991
American Airlines "RM generated additional $500 millions in revenue per year"

Time

Ultimate Super Saver Fares:

1) **Fare Restrictions**: Buy 30 days in advance, Saturday overnight, non-refundable.
2) **Capacity Control**: Restricted number of discount seats sold on each flight.

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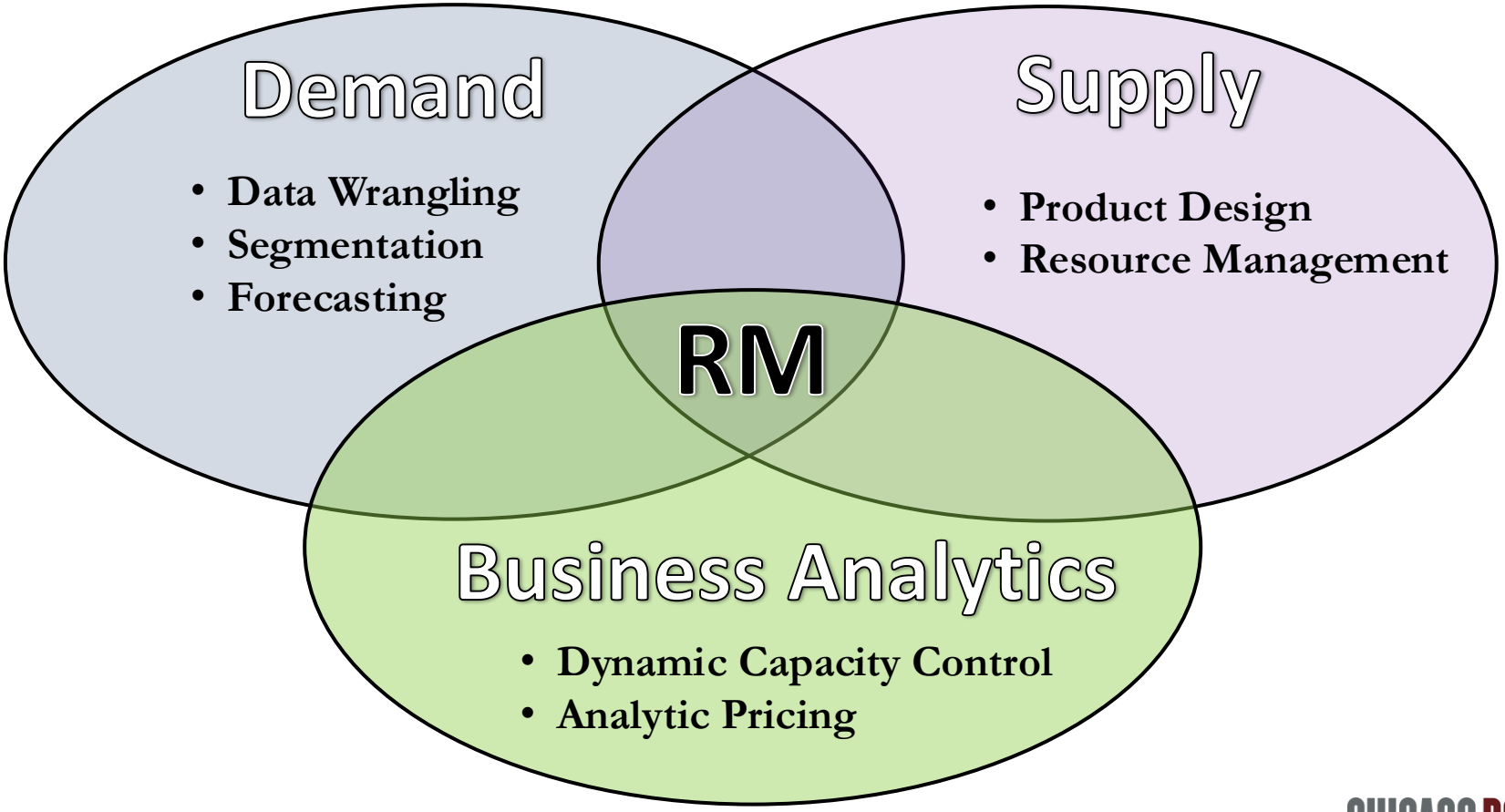
What is Revenue Management?

"Selling the right products to the right customers at the right prices and the right time through the right channel"



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Quick Assessment of RM Practice/Opportunity

FOUR LEVERS OF REVENUE MANAGEMENT	
PRICING BASIS	• Do you tend to charge the same price for your products or services to all customers? • Are your prices based mainly on costs and/or direct responses to competitor price movements?
INVENTORY ALLOCATION	• Do you sell your product or service on a first-come-first-served basis? • Do you charge the same price for your product or services during both high- and low-demand periods?
PRODUCT CONFIGURATION	• Does your product or service range require a wide variety of different resources to produce? • Does your product or service innovation usually involve radically new products?
DURATION CONTROL	• Do you have the same operating procedures during both busy and slow periods? • Do you accept variation in customer behavior as a given?

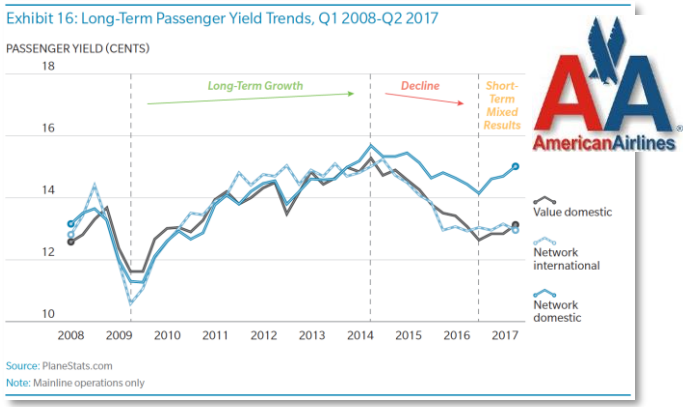
BUSINESS CONTEXT	
VARYING DEMAND	• Are there periods when you have too much capacity and periods when you have too little capacity? • Do you experience wide variation in customer demand across time periods?
PERISHABILITY	• Is it difficult to store your finished product or delay the availability of your service? • Are your business's outputs subject to loss of value over time?
CUSTOMER VARIATION	• Does your business serve a wide variety of customers who vary in their needs and willingness to pay for your product or service?

Source: Revenue Management Fundamentals (IMA)

Key performance indicator

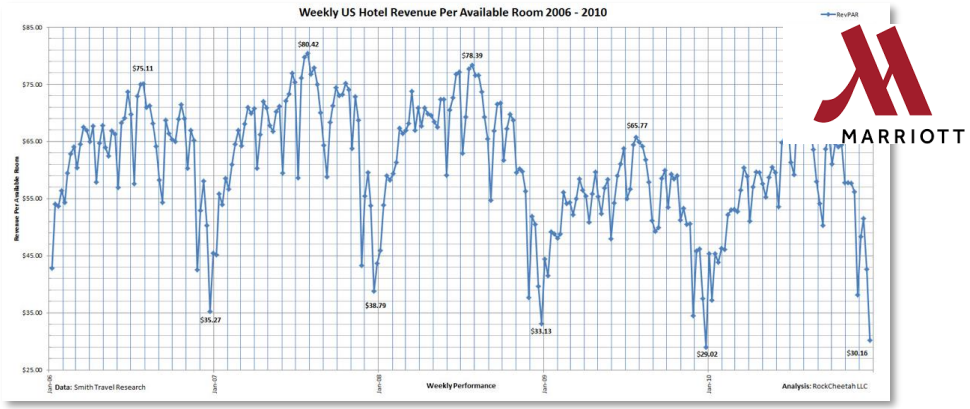
RevPAR = Revenue per Available Resource

RevPASM = Revenue per Available Seat Mile
= Yield × RPM

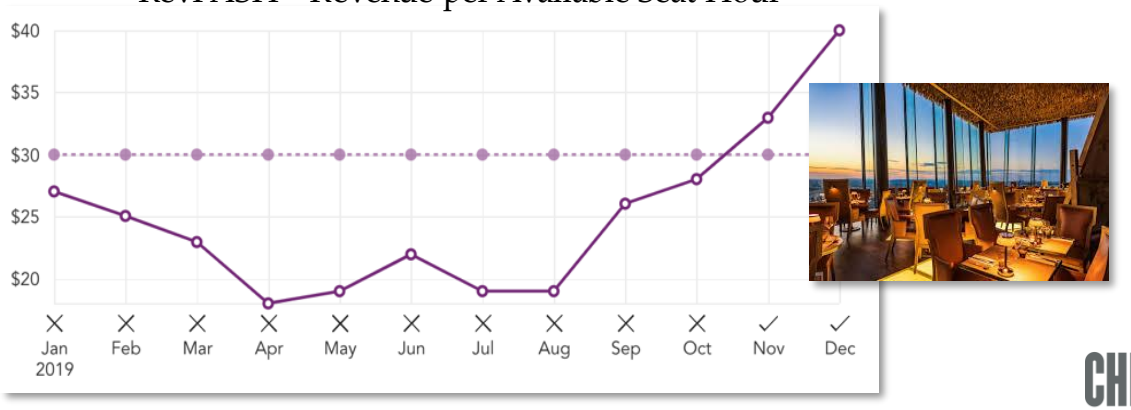


RevPAR = Revenue per Available Room
= ADR × OC

ADR = Average Daily Room Rate ($)
OC = Occupancy Rate (%)



RevPASH = Revenue per Available Seat Hour



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RM Holy Grail: Price Differentiation

SUPPLY:

Flight with 80 seats daily schedule Fixed Costs: Crew, fuel: $10000
Marginal costs: $0 (no-free-peanuts!)

DEMAND:

Business customers (B) have valuation $300 and market size 30.
Leisure customers (L) have valuation $100 and market size 80.

Best single price: _____ Maximum single price revenue: ____

Best RM strategy: Segment
Price at $300, $100
Control sales--- limit L sales to 50 (need restrictions!)
Revenue: _____

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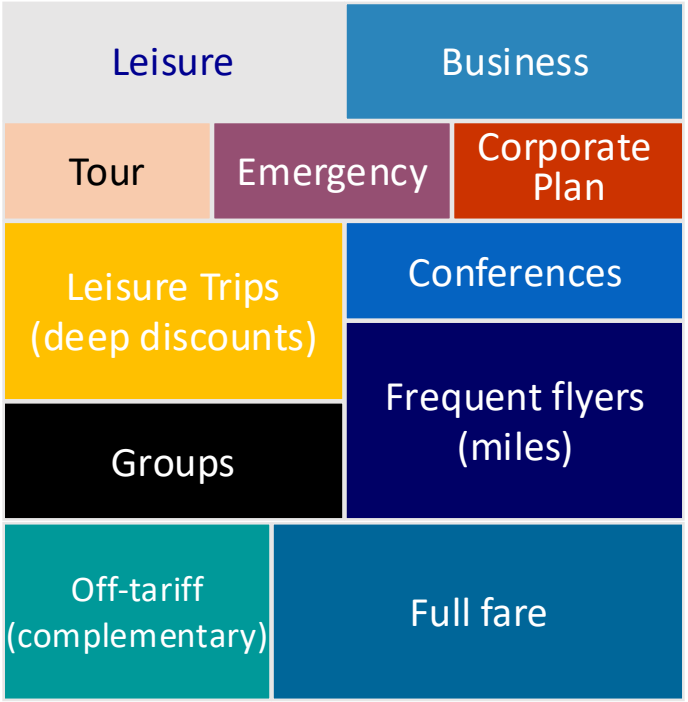
Implementation

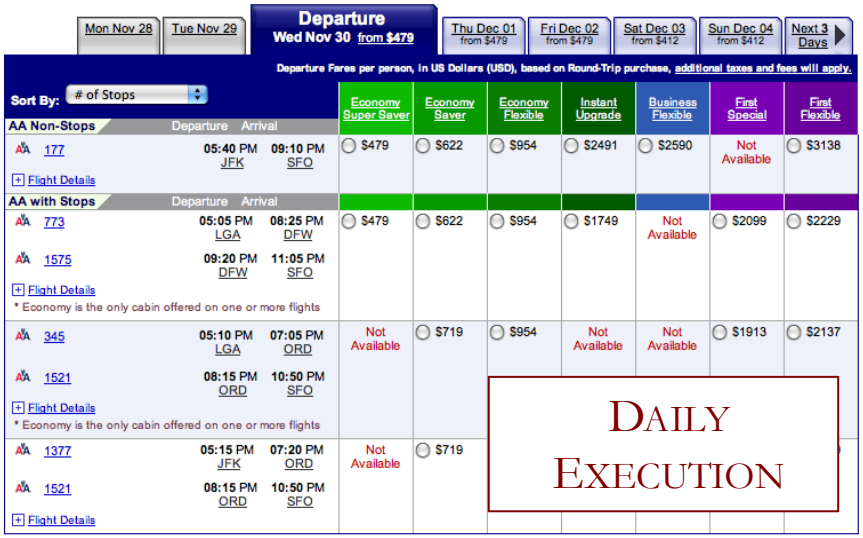
Customer Segmentation

Screening Devices



Product Design

Fences



Daily Execution

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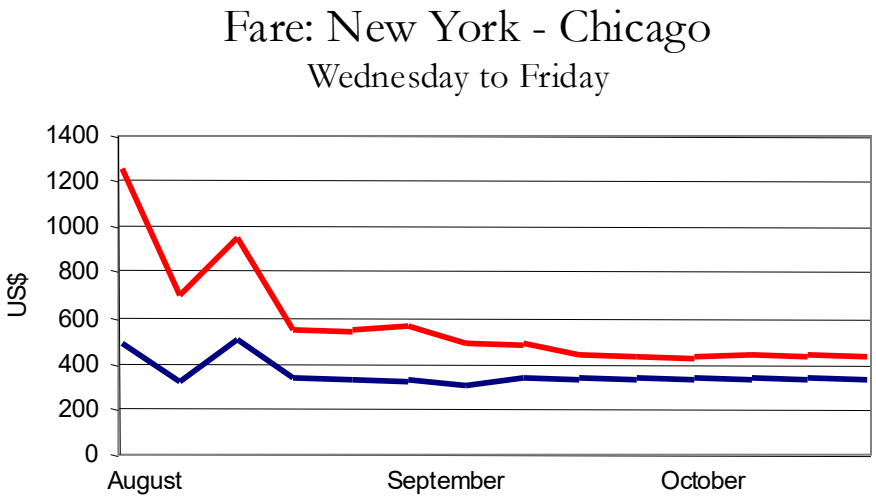
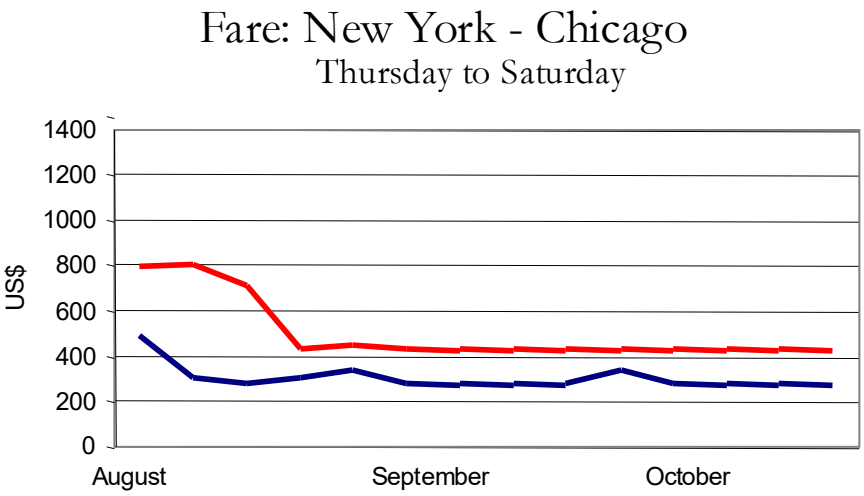
Revenue Management in Action





Wednesday-Friday fares are 15-20% higher than Thursday-Saturday fares!

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… but good judgment is needed!

[FreshDirect charges more in richer areas](#)

New York Post, 12/2013

Differentiated Pricing & Social Welfare

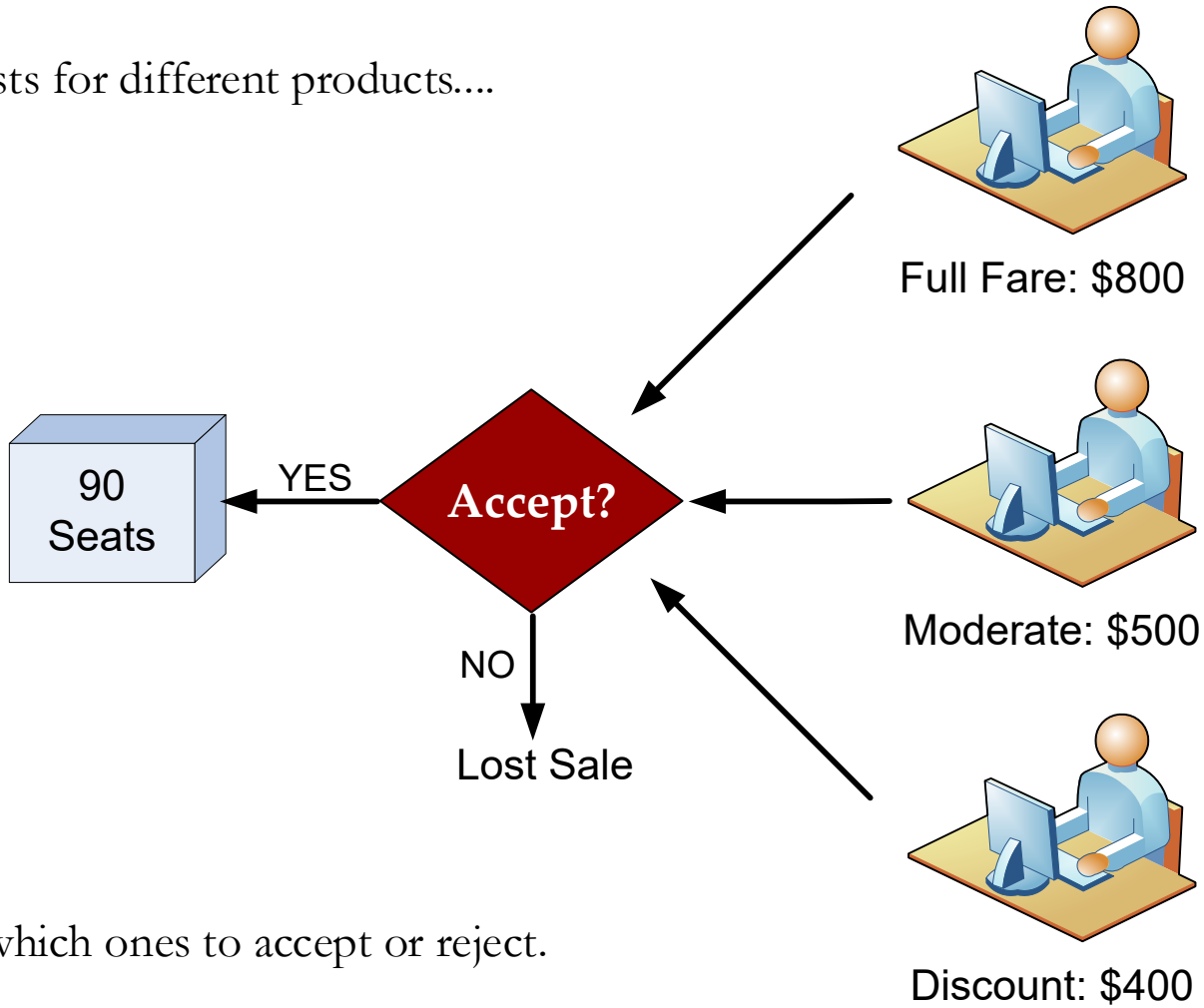
- Costly, $218B wasted in the US
- Externality and climate change
  - Methane 25x carbon GHG equivalent (IPCC)
  - US retailer vegetables gas emissions cause damage of $140M/year, $4.7B NPV
- Solutions
  - Regulations: Food Disposal Ban (MassDEP ’14, California’17)
  - Revenue Management: Dynamic Pricing

    Airline RM: wasted seats fell from 15% to 3%

Yield Management

Traditional airline/hotel quantity-based RM models:

Given requests for different products....



... decide which ones to accept or reject.

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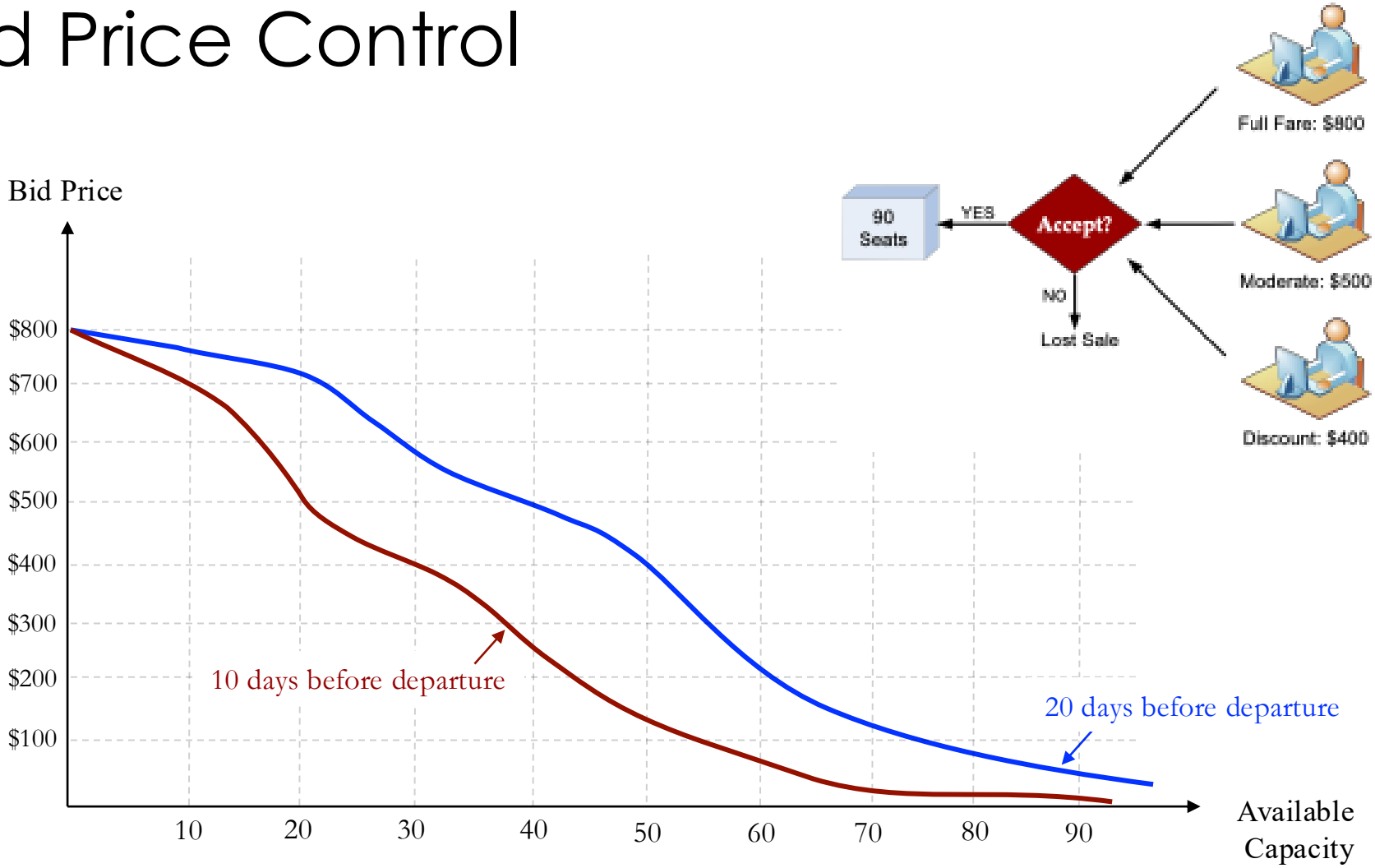
Bid Price Control



BID PRICES: Threshold price for "Accept/Reject" requests.

DISPLACEMENT COST: Expected revenue loss if we had one less unit of capacity.

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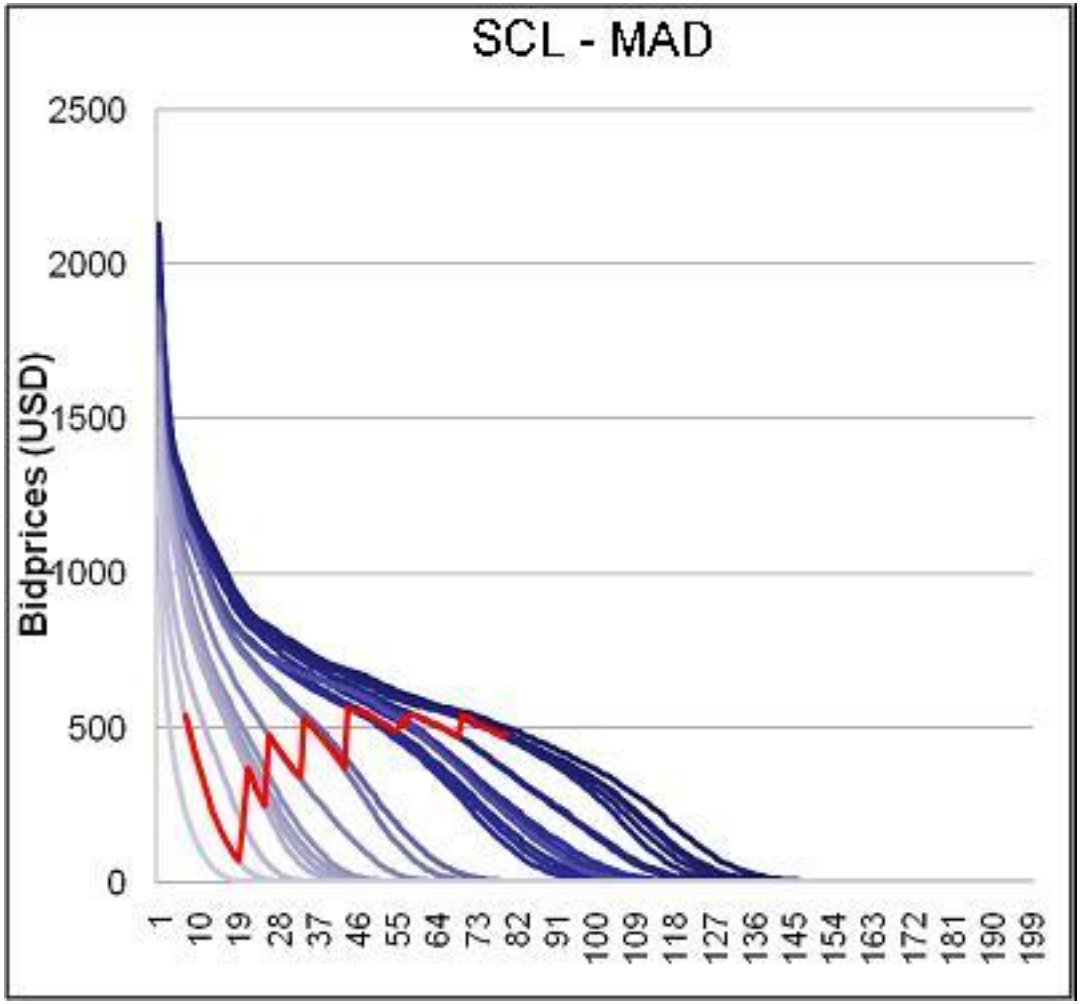
Example: Bid price control



Available Capacity (seats)

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Multiple Resources

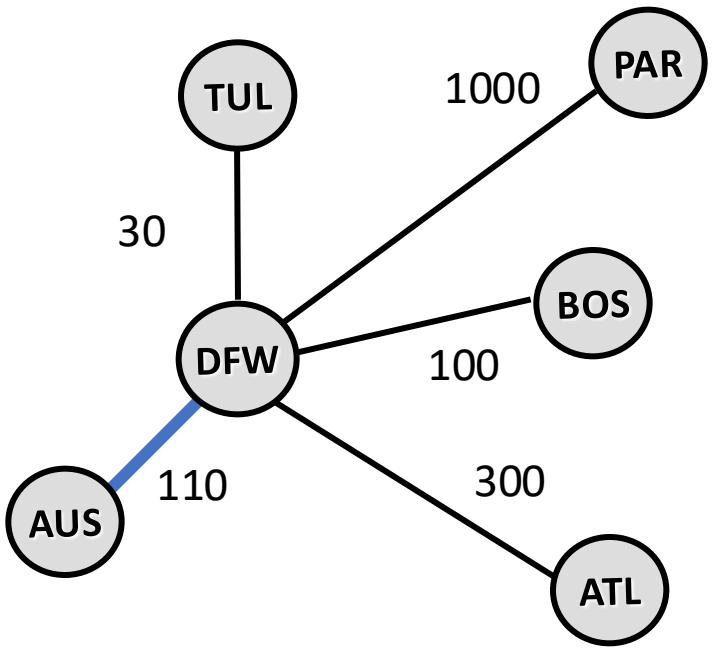
June 1st June 2nd June 3rd June 4th June 5th

Product-Resource Consumption Matrix

	Products															
	Single Night					Two Nights				Three Nights			Four Nights		Five Nights	Bid
Resources	Jun 1	Jun 2	Jun 3	Jun 4	Jun 5	Jun 1-2	Jun 2-3	Jun 3-4	Jun 4-5	Jun 1-3	Jun 2-4	Jun 3-5	Jun 1-4	Jun 2-5	Jun 1-5	Prices
Jun 1	1	0	0	0	0	1	0	0	0	1	0	0	1	0	1	150
Jun 2	0	1	0	0	0	1	1	0	0	1	1	0	1	1	1	150
Jun 3	0	0	1	0	0	0	1	1	0	1	1	1	1	1	1	90
Jun 4	0	0	0	1	0	0	0	1	1	0	1	1	1	1	1	150
Jun 5	0	0	0	0	1	0	0	0	1	0	0	1	0	1	1	180
Price ($)	160	150	120	150	180	300	250	250	310	400	400	420	500	550	700	
Bid Prices	150	150	90	150	180	300	240	240	330	390	390	420	540	570	720	
Status	Open	Open	Open	Open	Open	Open	Open	Open	Close	Open	Open	Open	Close	Close	Close	

BID PRICE OF A RESOURCE: Opportunity cost assigned to a resource.
BID PRICE OF A PRODUCT: Sum of the bid prices of the resources consumed by the product.
BID PRICE CONTROL: Accept a request if the price (fare) exceeds the Bid Price of the product.

Example: Bid price control



Org	Dst	Class	Fare	Bid Price
AUS	PAR	Y	$2,000	$1,110
AUS	BOS	Y	$1,500	$210
AUS	PAR	B	$1,000	$1,110
AUS	ATL	Y	$800	$410
AUS	BOS	B	$750	$210
AUS	ATL	B	$400	$410
AUS	TUL	Y	$300	$140
AUS	DFW	Y	$200	$110
AUS	TUL	B	$150	$140
AUS	DFW	B	$100	$110

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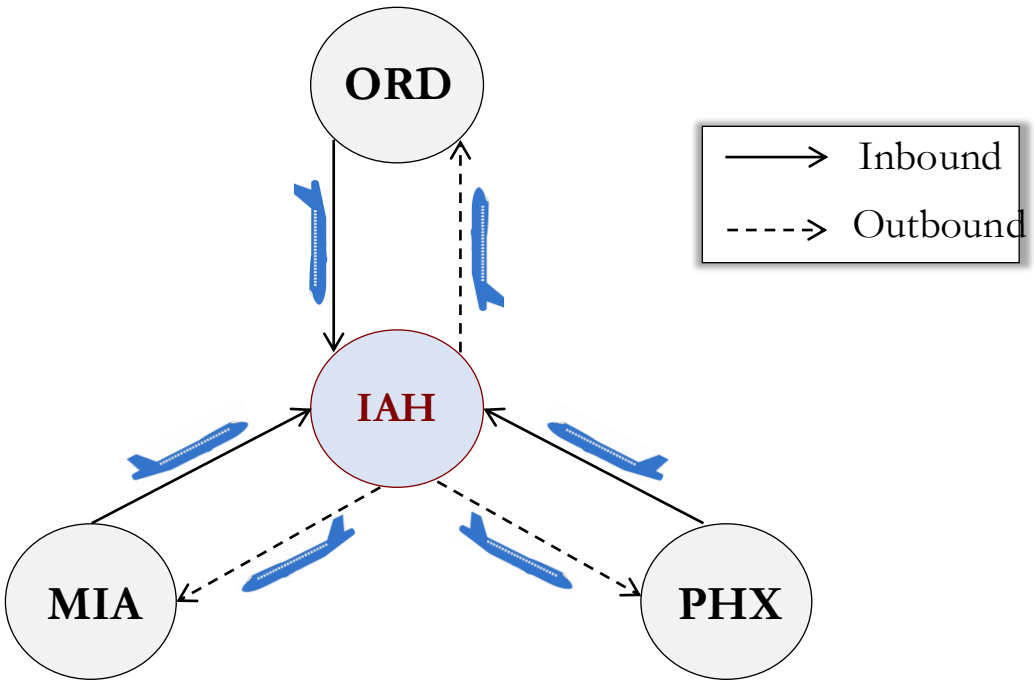
Computing Bid Prices



Data:

- Six resources (flights) with capacity 240 seats each.

Price for each Passenger Route

Origin	Destination: Houston	Chicago	Miami	Phoenix
Houston	-	$197	$110	$125
Chicago	$190	-	$282	$195
Miami	$108	$292	-	$238
Phoenix	$110	$192	$230	-

Demand During one Bank

		Destination: Houston	Chicago	Miami	Phoenix	Total Demand for flight to hub	
Origin	Houston	0	123	80	110	-	
	Chicago	130	0	98	88	316	(C-H)
	Miami	72	105	0	68	245	(M-H)
	Phoenix	115	90	66	0	271	(P-H)
Total demand for flight to spoke		-	318 (H-C)	244 (H-M)	266 (H-P)		

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Optimal Resource Allocation
(Solving Linear Programs)

- **Inputs**
  - f_i: Price (fare) of product (bundle) i
  - C_j: Available capacity of resource j
  - D_i: Average demand of product i
  - a_{ij}: Number of units of resource j consumed by one unit of product i.

- **Decision Variables**

  x_i: How many units of product i to sell.
  (partition booking limits)

- **Constraints**
  - Resource Capacity
  - Product Demand

- **Objective Function**

  Revenue Maximization

Objective: Max $\sum_i f_i x_i$

subject to

Resource Capacity:	$\sum_i a_{ij} x_i \le C_j$	(for all resources j)
Product Demand:	$x_i \le D_i$	(for all products i)
Nonngativities:	$x_i \ge 0$	

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Implementation: Bid Price Control

Objective: Max $\sum_i f_i x_i$

subject to

Resource Capacity: $\sum_i a_{ij} x_i \le C_j$ (for all resources j)

Product Demand: $x_i \le D_i$ (for all products i)

Nonngativities: $x_i \ge 0$

1. Solve linear program.
2. From the sensitivity report get shadow prices (bid price) for resource constraints: λ_j
3. For each product compute opportunity cost (bid price): $\pi_i = \sum_j a_{ij} \lambda_j$
4. Admission Rule: Accept demand for product i if and only if:

$$\boxed{f_i \ge \pi_i}$$

5. Re-optimize regularly (as new demand and data gets available). As a result, bid prices are not static but functions of time: $\pi(t)$

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Computing Bid Prices

Data:

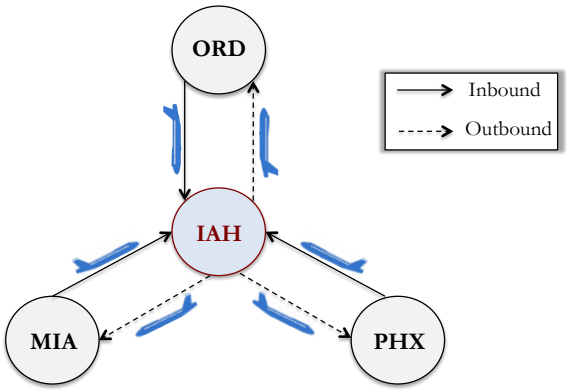
- Six resources (flights) with capacity 240 seats each.

Price for each Passenger Route

Origin	Destination: Houston	Chicago	Miami	Phoenix
Houston	-	$197	$110	$125
Chicago	$190	-	$282	$195
Miami	$108	$292	-	$238
Phoenix	$110	$192	$230	-

Demand During One Bank

		Destination: Houston	Chicago	Miami	Phoenix	Total Demand for flight to hub	
Origin	Houston	0	123	80	110	-	
	Chicago	130	0	98	88	316	(C-H)
	Miami	72	105	0	68	245	(M-H)
	Phoenix	115	90	66	0	271	(P-H)
Total demand for flight to spoke		-	318 (H-C)	244 (H-M)	266 (H-P)		



Decision Variables:

X_i: Number of Tickets Sold on each Product

For example, X_{MC} = # seats allocated to Miami-Chicago

Objective:

Max Revenues = $\$197 * X_{HC} + \$110 * X_{HM} + \ldots\ldots + \$230 * X_{PM}$

Constraints:

- Resource Capacity: (six constraints)

  $X_{CH} + X_{CM} + X_{CP} \leq 240$ (Flight Chicago-Houston)

  ……..

  $X_{HP} + X_{CP} + X_{MP} \leq 240$ (Flight Houston-Phoenix)

- Product Demand: (twelve constraints)

  $X_{MC} \leq 105$ (Demand Miami-Chicago)

  … …..

  $X_{CP} \leq 88$ (Demand Chicago-Phoenix)

- Non-negativities: (twelve constraints) $X \geq 0$

- Decision Variables
- Constraints
- Objective Function

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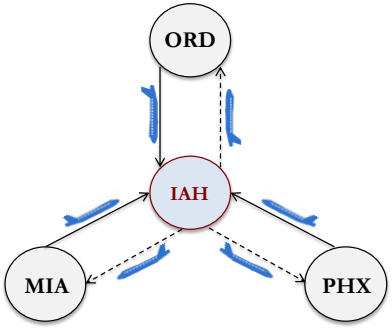
Optimal Resource Allocation
(Required Inputs)



	HC	HM	HP	CH	CM	CP	MH	MC	MP	PH	PC	PM
Contribution	197	110	125	190	282	195	108	292	238	110	192	230

Product-Resource Consumption Matrix

	HC	HM	HP	CH	CM	CP	MH	MC	MP	PH	PC	PM	Capacity	Bid Price
Flight CH	0	0	0	1	1	1	0	0	0	0	0	0	240	190
Flight MH	0	0	0	0	0	0	1	1	1	0	0	0	240	100
Flight PH	0	0	0	0	0	0	0	0	0	1	1	1	240	0
Flight HC	1	0	0	0	0	0	0	1	0	0	1	0	240	192
Flight HM	0	1	0	0	1	0	0	0	0	0	0	1	240	92
Flight HP	0	0	1	0	0	1	0	0	1	0	0	0	240	5

Product-Demand Matrix

	HC	HM	HP	CH	CM	CP	MH	MC	MP	PH	PC	PM	Forecast	Shadow Price
Demand HC	1	0	0	0	0	0	0	0	0	0	0	0	123	5
Demand HM	0	1	0	0	0	0	0	0	0	0	0	0	80	18
….	….	….					….	….			….	….	….	….
Demand MH	0	0	0	0	0	0	1	0	0	0	0	0	72	8
….	….	….					….	….			….	….	….	….
Demand PM	0	0	0	0	0	0	0	0	0	0	0	1	66	138

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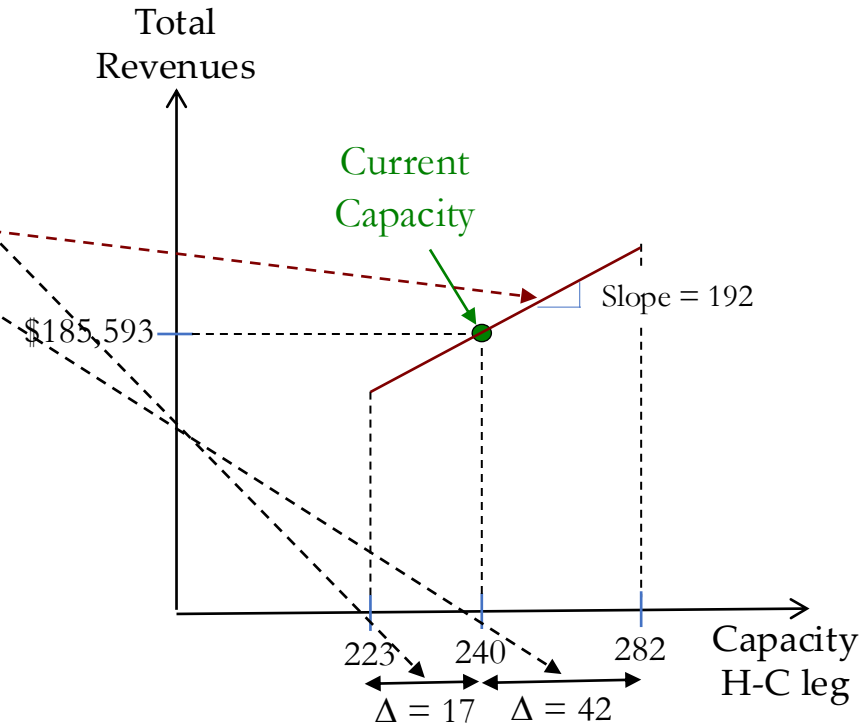
Excel Solution: Sensitivity Report

Capacity (seats) Allocation

		Destination			
		Houston	Chicago	Miami	Phoenix
Origin	Houston	0	123	80	110
	Chicago	84	0	94	62
	Miami	72	100	0	68
	Phoenix	115	17	66	0

Excel Sensitivity Report

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
D44	H-C Capacity Used	240	192	240	42	17
D45	H-M Capacity Used	240	92	240	4	46
D46	H-P Capacity Used	240	5	240	26	46
D47	C-H Capacity Used	240	190	240	46	84
D48	M-H Capacity Used	240	100	240	5	42
D49	P-H Capacity Used	198	0	240	1E+30	42
D55	HC Tickets Sold	123	5	123	17	42
D56	HM Tickets Sold	80	18	80	46	4
D57	HP Tickets Sold	110	120	110	46	26
D58	CH Tickets Sold	84	0	130	1E+30	46
D59	CM Tickets Sold	94	0	98	1E+30	4
D60	CP Tickets Sold	62	0	88	1E+30	26
D61	MH Tickets Sold	72	8	72	42	5
D62	MC Tickets Sold	100	0	105	1E+30	5
D63	MP Tickets Sold	68	133	68	42	5
D64	PH Tickets Sold	115	110	115	42	115
D65	PC Tickets Sold	17	0	90	1E+30	73
D66	PM Tickets Sold	66	138	66	42	4



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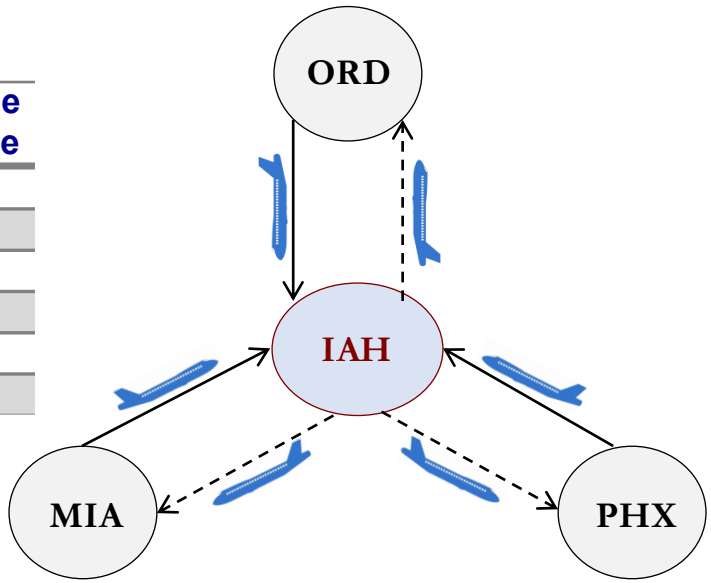
Implementation: Bid Price Control

Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
H-C	240	192	240	42	17
H-M	240	92	240	4	46
H-P	240	5	240	26	46
C-H	240	190	240	46	84
M-H	240	100	240	5	42
P-H	198	0	240	1E+30	42



Implementation:

1. We receive a request for a Miami – Chicago for $300. Should we accept it?

2. We receive a request for a Chicago–Miami for $250. Should we accept it?

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Key Ideas

- Services: Unique challenges and opportunities
  - The human factor: managing perceptions and expectations.
  - Closing the gaps: No news is NOT good news
  - Ops challenges: Intangible, Perishable, Heterogeneous, Simultaneous
  - Ops opportunities: customer-operator, standardize to customize

- Service Oriented Strategies
  - Service Materialization
  - Capacity Management: Matching Supply and Demand
  - Segmentation and Differentiation
  - Choosing the right strategy: Assembly line, self-service, personal attention

- Revenue Management of Perishable Assets
  - The Holy Grail: differentiation customers and products
  - Tactics for Revenue Management
    - Demand Segmentation & Screening Devices
    - Product Versioning: Fences & Incentive Compatibility
  - Matching the Right customer with the Right product at the Right time
    - Bid Price Controls & Marginal Value of Resources

After Class Activities

- **INDIVIDUAL:**

  TAKEAWAY #3:

  Contribute a second individual idea for final group project, similar to takeaway 1. (1 paragraph)

- **GROUP:**

  CASE: SOF-OPTICS, INC.

  Managing process variability.